

performance on complex tasks again exhibiting threshold-like behavior [5]. While specific architectural details remain undisclosed, GPT-4's results anchor the empirical end of the scaling continuum, showing that emergent abilities persist and strengthen as models grow.

Wei et al. systematically formalized the concept of emergent abilities, defining them as capabilities that cannot be linearly extrapolated from smaller models' performance [6]. By analyzing hundreds of evaluation points across model families—GPT-3, Gopher, Chinchilla, PaLM—they plotted accuracy versus log-scale model size and observed sigmoidal or step-like patterns instead of smooth power-law gains. These discontinuities appeared in arithmetic, translation, reasoning, and word-sense disambiguation. Importantly, Wei et al. emphasized that emergent abilities reflect abrupt changes in task performance, not necessarily smooth loss improvements. This observation directly motivated our focus on external behavioral metrics—such as dispersion and vocabulary composition—that reveal reorganizations hidden in standard loss curves. However, Wei et al. left the underlying mechanism open, describing emergence primarily as an empirical phenomenon tied to scale.

E. QUESTIONING THE REALITY OF EMERGENT ABILITIES

Schaeffer, Miranda, and Koyejo challenged the interpretation of emergent abilities as evidence of fundamental phase transitions [20]. They argued mathematically and empirically that many step-like curves reported in Wei et al. arise from discrete, non-linear evaluation metrics rather than abrupt internal reorganizations. When an underlying performance variable—such as the probability of correct reasoning—increases smoothly with scale, threshold-based metrics like binary accuracy can still produce the illusion of sudden jumps. By re-plotting results using continuous metrics such as cross-entropy or expected loss, Schaeffer et al. found mostly smooth scaling laws across model families. They concluded that observed “emergence” may often be an artifact of binarized or non-linear task scoring, though they acknowledged that some genuine discontinuities could exist and encouraged mechanistic probes of internal representations.

This critique is foundational for our study. Schaeffer et al. demonstrate that not every S-curve or cusp in task performance implies a physical-like phase transition, underscoring the necessity of continuous, internal diagnostics rather than external accuracy thresholds. Our use of Poisson and sub-Poisson statistics, KL divergence, and vocabulary composition directly addresses this concern: these are continuous measures of linguistic structure that evolve throughout training, independent of arbitrary task cutoffs. By detecting synchronized discontinuities across multiple continuous probes—dispersion, KL divergence, word length, vocabulary dynamics—we provide converging evidence for a genuine reorganization rather than a metric artifact. In this sense, our approach heeds Schaeffer et

al.'s call for mechanistic, continuous analysis while still identifying discontinuous behavior.

F. GROKING AND DISCONTINUOUS TRAINING DYNAMICS

The phenomenon of grokking, introduced by Power et al., provides the most direct small-scale analogue to our findings [16]. Training small transformers and MLPs on modular-arithmetic datasets, Power et al. observed that models first overfit—achieving near-zero training loss but random test accuracy—and then, after many additional epochs, suddenly transition to perfect generalization. This delayed, abrupt reorganization occurs despite smooth loss curves and depends on regularization strength (weight decay) and dataset diversity. Visualizations of embedding geometry reveal that during grokking, learned features rotate and align with the true algorithmic structure, suggesting the model moves from a memorization basin to a generalization basin in weight space. Grokking thus establishes that neural networks, even at modest scale, can remain in a long plateau before abruptly reorganizing internally—a discontinuity not captured by training loss alone.

Agarwal, Morcos, and Krishnamurthy extended this work through mechanistic interpretability, applying singular value decomposition, neuron activation analysis, and embedding alignment to track how grokking unfolds inside the network [17]. They introduced quantitative progress measures capturing the degree of algorithmic abstraction in weights and representations. Their key finding is that even when external accuracy remains flat, internal circuits gradually form compositional structures; once these structures cross a stability threshold, generalization jumps. This threshold coincides with sudden drops in representational entropy and changes in gradient flow, supporting the view that grokking is an internal reorganization event, not gradual refinement. For our work, Agarwal et al. provide a mechanistic complement to our external metrics: both reveal latent buildup followed by rapid consolidation. The parallel is direct—our dispersion and KL diagnostics capture analogous reorganizations in linguistic structure, where fragments coalesce into coherent words at a critical training epoch.

Dolev et al. documented similar discontinuities in neural machine translation, where performance on morphologically rich and low-resource languages leaps suddenly once model or dataset size crosses a threshold [18]. Using Transformer architectures trained on WMT datasets, they observed that while translation quality grows smoothly at small scales, it jumps sharply beyond a critical point. They attributed the discontinuity to improved contextual representation and capacity to memorize rare-word alignments, terming these “critical advantages” of scaling. Although Dolev et al. did not analyze internal representation dynamics or apply statistical probes, their empirical curves parallel the abrupt lexical consolidation we observe, reinforcing the idea that neural sequence models can display threshold-like improvements in linguistic coherence.

G. THEORETICAL FOUNDATIONS: GROKING AS A FIRST-ORDER PHASE TRANSITION

Rubin, Seroussi, and Ringel provided the first formal statistical-physics model of grokking, directly linking training dynamics to phase transitions [19]. They studied a two-layer teacher–student network trained on modular addition and derived an effective potential describing the system’s macroscopic state. The potential exhibits two competing minima: one corresponding to memorization (fitting training data without generalization) and another to generalization (capturing the underlying rule). As regularization strength or training time increases, the minima exchange stability, producing a discontinuous jump in the order parameter—identifying grokking as a first-order phase transition. Rubin et al. supported this mapping through analytical mean-field approximations and numerical simulations, demonstrating hysteresis and mixed phases characteristic of first-order transitions. They also identified intermediate regimes with Gaussian mixture features, matching empirical embedding structures seen in earlier grokking studies.

This theoretical work establishes the strongest justification for describing discontinuities in neural training as genuine phase transitions. Rubin et al. demonstrate rigorously that smooth local updates—gradient descent steps—can yield abrupt, global changes in representational order once competing learning modes exchange stability. Although their model applies to simplified two-layer networks and algorithmic tasks, the conceptual framework extends naturally to our setting. The dispersion flip we observe—correct words transitioning from near-Poisson to sub-Poisson while incorrect words move from sub-Poisson to Poisson-like—can be interpreted as the system crossing a critical threshold where the “generalization minimum” (coherent word usage) overtakes the “memorization minimum” (fragmentary output). In this view, our Poisson-centered diagnostics serve as order parameters tracking which representational mode dominates as training progresses.

H. POSITIONING OUR WORK

Against this backdrop, our study occupies a distinct and underexplored position. Most empirical investigations of emergent abilities (Wei et al., Brown et al., Chowdhery et al.) focus on billion-parameter models evaluated after training, often requiring logarithmic rescaling of compute or model size to reveal discontinuities. In contrast, we investigate a small, 3.6M-parameter transformer and track its evolution directly along the linear training axis, demonstrating that phase-transition-like reorganizations are observable in modest systems without log-scaling. Grokking research (Power et al., Agarwal et al.) provides the closest analogue—small models, abrupt reorganization—but focuses on algorithmic datasets and generalization accuracy. We extend this line to linguistic tasks, introducing Poisson-based statistical probes that detect discontinuities in vocabulary structure, word length, and dispersion rather than task accuracy alone.

Theoretically, we leverage the frameworks of Bahri et al.,

Miller et al., and especially Rubin et al. to interpret our findings. The dispersion flip—correct words shifting from near-Poisson to sub-Poisson, incorrect words from sub-Poisson to Poisson-like—mirrors the competition between memorization and generalization minima in Rubin et al.’s effective potential. Our KL divergence and vocabulary dynamics corroborate this interpretation, showing synchronized cusps and collapses coinciding with the onset of multi-character word coherence. Methodologically, we heed Schaeffer et al.’s caution by employing continuous, internal metrics rather than binary task thresholds, thereby avoiding metric-induced artifacts while still identifying genuine discontinuities.

In sum, our contribution is threefold: (1) we demonstrate that phase-transition-like reorganizations are not unique to large language models but arise in small transformers trained on character-level linguistic data; (2) we show that such transitions can be detected directly in linear training space using Poisson-centered diagnostics, without requiring log-scaling or massive compute; and (3) we establish that these transitions occur surprisingly early in training as lexical coherence emerges, with word-level structure serving as a natural order parameter. By bridging statistical physics, emergent abilities research, and small-model training dynamics, we provide both empirical evidence and interpretive tools for understanding nonlinear reorganizations in neural language learning.

III. METHODS

A. MODEL AND DATA

We trained a compact character-level transformer designed to resemble the basic architecture of larger language models while remaining computationally controllable. The architecture consists of an embedding dimension of 192, 8 transformer layers, and 6 attention heads (approximately 3.6M parameters) with a context length of 128 characters. The model was trained on the Tiny Shakespeare corpus, which contains ~ 1.1 M character tokens that span 65 unique characters.

To assess robustness, we ran 5 independent seeds for 0–600 epochs. At each checkpoint, we sampled 30,000 tokens using a fixed decoding configuration (temperature $T = 1.0$, greedy top-1 unless otherwise noted).

B. SEGMENTATION AND CORRECTNESS LABELING

Generated text was segmented into words using whitespace and punctuation boundaries. We then classified each unit as either:

- **Correct:** if it appears in the corpus vocabulary.
- **Incorrect:** if it does not appear in the corpus vocabulary.

This operational split provides a way to separate the legitimate vocabulary from exploratory or erroneous output. Although it excludes words beyond the training distribution, this operational definition still captures the model’s progression toward vocabulary-level structure, thereby offering a practical basis for experimental analysis.

C. VOCABULARY GROWTH AND WORD LENGTH

To capture the linguistic impact of the transition, we track two complementary measures that quantify the model's progression from fragmentary output to compositional word formation.

Unique vocabulary counts ($V_{\text{uniq,correct}}$ and $V_{\text{uniq,incorrect}}$) measure the diversity of word types generated in each class. The unique correct vocabulary reflects the model's expanding repertoire of legitimate forms, while the unique incorrect vocabulary tracks the proliferation and eventual collapse of erroneous patterns. By monitoring both separately, we can identify when the model shifts from exploring fragmentary errors to consolidating valid words. A peak in incorrect vocabulary diversity followed by collapse signals that the model has exhausted exploratory combinations and committed to structured patterns.

Average word length ($\bar{L}$) provides a direct measure of compositional complexity:

$$\bar{L} = \frac{1}{N_{\text{correct}}} \sum_{w \in \text{correct}} \text{len}(w). \quad (1)$$

Early in training, when the model primarily generates single-character outputs or short two-letter fragments, $\bar{L}$ remains near 1.5 characters. As the model learns to compose longer, multi-character words, $\bar{L}$ increases. A sharp, S-shaped rise in average word length signals a qualitative shift from character-level fragmentation to word-level composition. This metric serves as a natural linguistic correlate of the dispersion transition: sub-Poisson regularity in correct-word usage emerges precisely when the model begins producing stable, multi-character forms.

Together, these measures reveal how the system reorganizes from producing short, error-prone fragments to longer coherent words. The synchronized dynamics—incorrect vocabulary collapse, correct vocabulary acceleration, and word-length increase—provide converging evidence that the statistical reorganization captured by dispersion and KL divergence manifests as a tangible linguistic restructuring.

D. POISSON BASELINE, DISPERSION, AND SNAPSHOTS

To test whether words appear randomly or in a structured way, we divided generated text into non-overlapping windows of $W = 21$ words. If word occurrences were independent random events, the number of words N per window would follow a Poisson distribution:

$$P(N = n; \lambda) = \frac{\lambda^n e^{-\lambda}}{n!}, \quad \text{Var}[N] = \lambda, \quad (2)$$

with mean λ equal to the variance.

Deviations from this Poisson baseline are summarized by the **index of dispersion**:

$$D = \frac{\sigma^2}{\mu}, \quad (3)$$

where μ and σ^2 are the mean and variance of the count distribution.

- $D = 1$: **Poissonian** — events occur randomly and independently, with variance equal to the mean. This represents the natural benchmark of uncorrelated randomness.
- $D < 1$: **Sub-Poissonian** — variance is suppressed relative to the mean. Events are more evenly spaced than chance would predict, signaling hidden order or regularity. In physics, this regime is associated with quantum antibunching of photons.
- $D > 1$: **Super-Poissonian** — variance exceeds the mean. Events occur in bursts or clusters, as seen in chaotic or heavy-tailed processes.

In our linguistic setting, these regimes carry intuitive interpretations.

- **Correct words** migrate from near-Poisson to sub-Poisson as training progresses, reflecting a shift from rare, scattered successes to stable, regular usage patterns.
- **Incorrect words** begin in a repetitive, variance-suppressed regime (sub-Poisson) dominated by short, recurring fragments like “th” or “yo,” then disperse randomly as errors diminish, drifting toward $D \approx 1$ as they become sparse, independent noise.

To illustrate these dynamics, Figs. 1–3 show side-by-side histograms of incorrect (left) and correct (right) word counts at representative checkpoints. Each panel displays the empirical distribution (blue bars) overlaid with the fitted Poisson baseline (orange dots).

Step 0 (Early training): Incorrect words show sub-Poisson clustering, with narrow distributions reflecting repetitive short fragments. Correct words are extremely sparse and scattered, displaying near-Poisson randomness with very few successful multi-character forms.

Step 300 (Mid-training, approaching transition): Incorrect words maintain sub-Poisson clustering with narrow, variance-suppressed distributions as repetitive fragments persist. Correct words begin to show emerging structure but remain scattered and near-Poisson, indicating the model is starting to discover valid patterns but usage remains irregular.

Step 599 (Late training, post-transition): The dispersion flip is complete. Incorrect words have broadened to near-Poisson distributions, consistent with sparse, independent residual errors that occur randomly rather than in repetitive clusters. Correct words have contracted into tight, sub-Poisson distributions with suppressed variance, reflecting regular, structured usage where coherent words appear predictably and consistently.

This visual progression directly illustrates the dispersion flip that defines the transition: incorrect words evolve from sub-Poisson (repetitive clustered fragments like “th, th, th”) to Poisson (sparse random errors), while correct words evolve from near-Poisson (scattered rare successes) to sub-Poisson (regular coherent usage).